

Power BI Assignment 2 – DAX & Data Visualization

Calculated Columns:

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Steps to Create Calculated Column

1. Go to **Data View** in Power BI
2. Select the **Order Details** table
3. Click on **New Column**
4. Enter the following DAX formula:

Category Type =

'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing - Hankerchief	₹ 24	Below Average
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing - Hankerchief	₹ 1,285	Below Average
B-25615	₹ 68	20	5	Clothing	Hankerchief	0.294117647	Profit	Clothing - Hankerchief	₹ 340	Below Average
B-25616	₹ 42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing - Hankerchief	₹ 210	Below Average
B-25624	₹ 26	12	3	Clothing	Hankerchief	0.461538462	Profit	Clothing - Hankerchief	₹ 78	Below Average
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing - Hankerchief	₹ 194	Below Average
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120	Below Average
B-25638	₹ 154	39	3	Clothing	Hankerchief	0.253246753	Profit	Clothing - Hankerchief	₹ 462	Below Average
B-25654	₹ 34	12	3	Clothing	Hankerchief	0.352941176	Profit	Clothing - Hankerchief	₹ 102	Below Average
B-25656	₹ 6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6	Below Average
B-25656	₹ 56	18	2	Clothing	Hankerchief	0.321428571	Profit	Clothing - Hankerchief	₹ 112	Below Average
B-25670	₹ 24	1	2	Clothing	Hankerchief	0.041666667	Profit	Clothing - Hankerchief	₹ 48	Below Average
B-25670	₹ 14	2	1	Clothing	Hankerchief	0.142857143	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25730	₹ 18	8	2	Clothing	Hankerchief	0.444444444	Profit	Clothing - Hankerchief	₹ 36	Below Average
B-25751	₹ 32	7	3	Clothing	Hankerchief	0.21875	Profit	Clothing - Hankerchief	₹ 96	Below Average
B-25751	₹ 106	15	7	Clothing	Hankerchief	0.141509434	Profit	Clothing - Hankerchief	₹ 742	Below Average
B-25757	₹ 14	5	1	Clothing	Hankerchief	0.357142857	Profit	Clothing - Hankerchief	₹ 14	Below Average
B-25757	₹ 17	7	3	Clothing	Hankerchief	0.411764706	Profit	Clothing - Hankerchief	₹ 51	Below Average
B-25771	₹ 148	59	3	Clothing	Hankerchief	0.398648649	Profit	Clothing - Hankerchief	₹ 444	Below Average
B-25783	₹ 26	2	2	Clothing	Hankerchief	0.076923077	Profit	Clothing - Hankerchief	₹ 52	Below Average
B-25784	₹ 15	4	1	Clothing	Hankerchief	0.266666667	Profit	Clothing - Hankerchief	₹ 15	Below Average
B-25793	₹ 18	1	3	Clothing	Hankerchief	0.055555556	Profit	Clothing - Hankerchief	₹ 54	Below Average
B-25797	₹ 31	1	2	Clothing	Hankerchief	0.032258065	Profit	Clothing - Hankerchief	₹ 62	Below Average

Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

Steps to Create Calculated Column

1. Go to **Data View**
2. Select the **Order Details** table
3. Click **New Column**
4. Enter this DAX formula:

Revenue per Order =

`'Order Details'[Amount] * 'Order Details'[Quantity]`

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File Home Help Table tools

Name: Order Details

Manage relationships Relationships Calculations Calendars

Structure

Table: Order Details (1,500 rows)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing - Hankerchief	₹ 24	Below Average
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing - Hankerchief	₹ 1,285	Below Average
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B-25656	₹ 56	18	2	Clothing	Hankerchief	0.321428571	Profit	Clothing - Hankerchief	₹ 112	Below Average
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B-25797	₹ 31	1	2	Clothing	Hankerchief	0.032258065	Profit	Clothing - Hankerchief	₹ 62	Below Average

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Steps

1. Go to **Data View**
2. Select **Order Details** table
3. Click **New Column**
4. Enter the following DAX formula:

Sales Category =

IF(

`'Order Details'[Amount] >`

`CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details'))),`

`"Above Average",`

"Below Average"

)

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File Home Help Table tools

Name Order Details

Manage relationships New Quick measure measure column New table Mark as date table

Structure Relationships Calculations Calendars

Table: Order Details (1,500 rows)

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Calculated Measures:

- Calculate Order Count: Define a measure to count the total number of orders in the Order Details table.

✓ Steps to Create Measure

1. Go to **Report View** or **Data View**
2. Select the **Order Details** table
3. Click **New Measure**
4. Enter this DAX formula:

Order Count = DISTINCTCOUNT('Order Details'[Order ID])

Calculate Average Profit in Delhi: Create a measure to calculate the average profit for orders placed in Delhi.

Steps to Create the Measure

1. Go to **Report View**
2. Select your table (for example: **Order Details** or Orders table)
3. Click **New Measure**
4. Enter the following DAX formula:

□ DAX Formula

```
Average Profit Delhi =  
CALCULATE(  
    AVERAGE('Order Details'[Profit]),  
    'Order Details'[City] = "Delhi"  
)
```

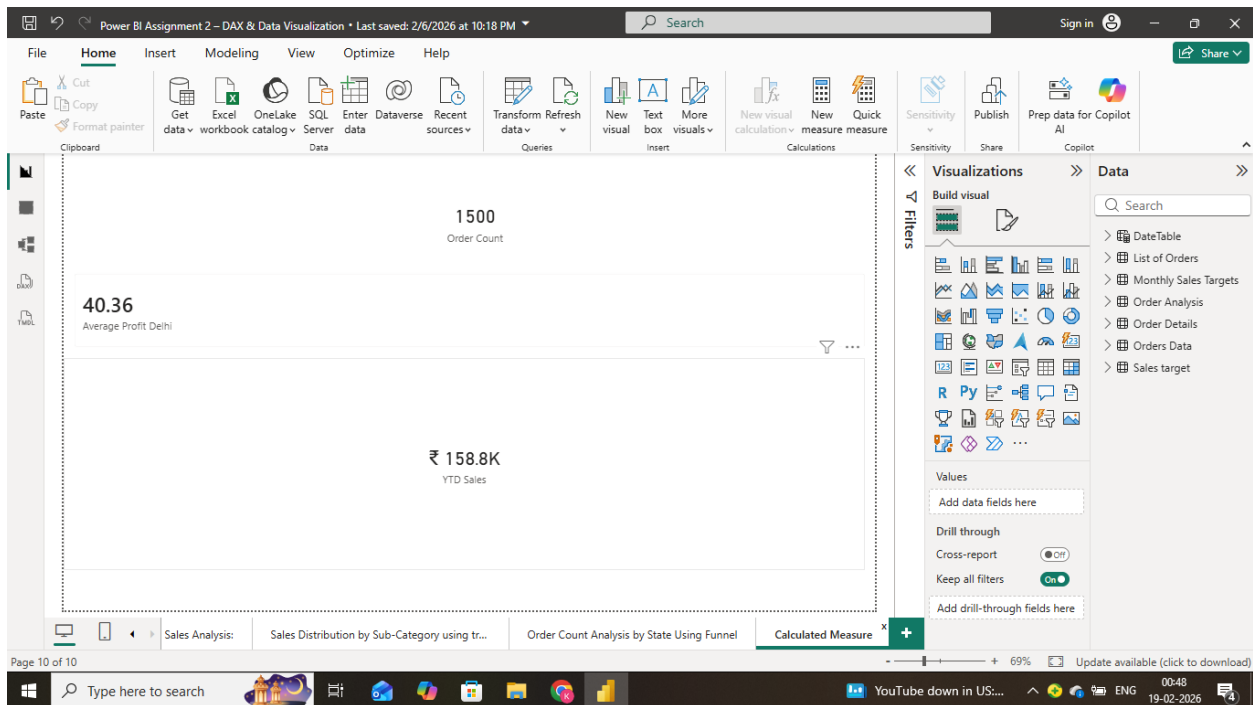
Calculate Year-to-Date (YTD) Sales: Define a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

Step 1: Create Total Sales Measure

```
Total Sales = SUM('Order Details'[Sales Amount])
```

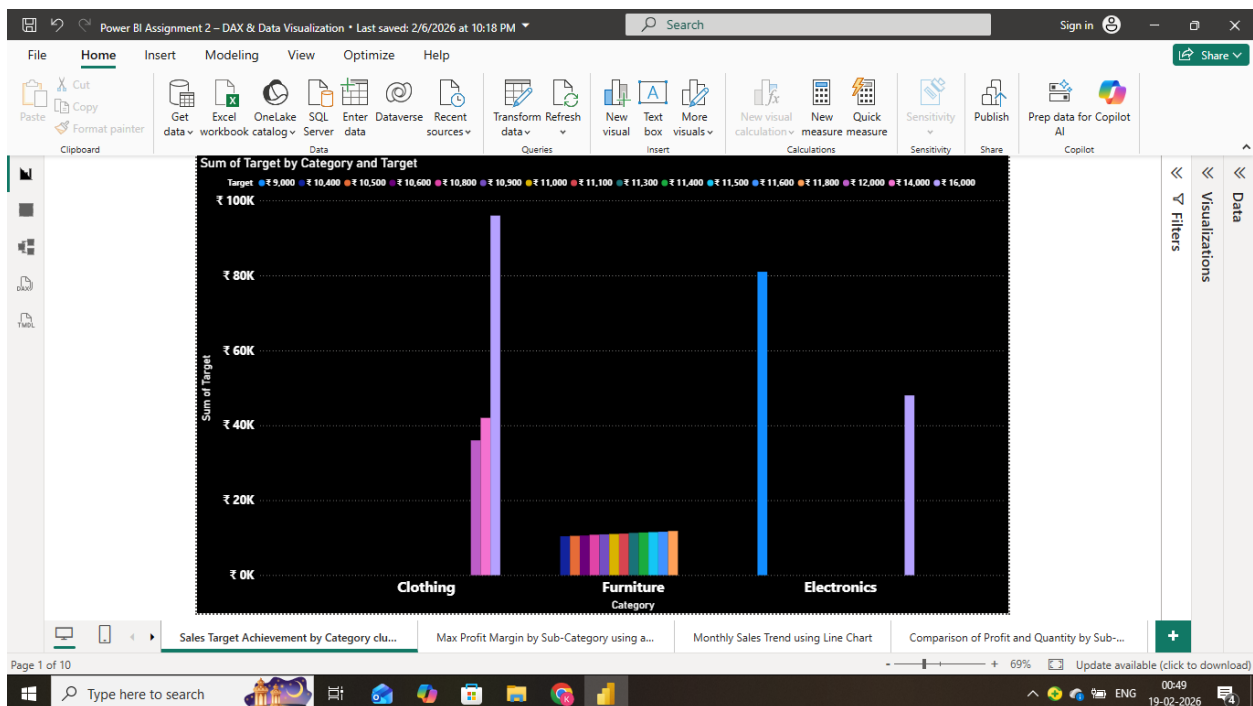
□ Step 2: Create YTD Sales Measure

```
YTD Sales =  
CALCULATE(  
    [Total Sales],  
    DATESYTD('Date Table'[Date])  
)
```



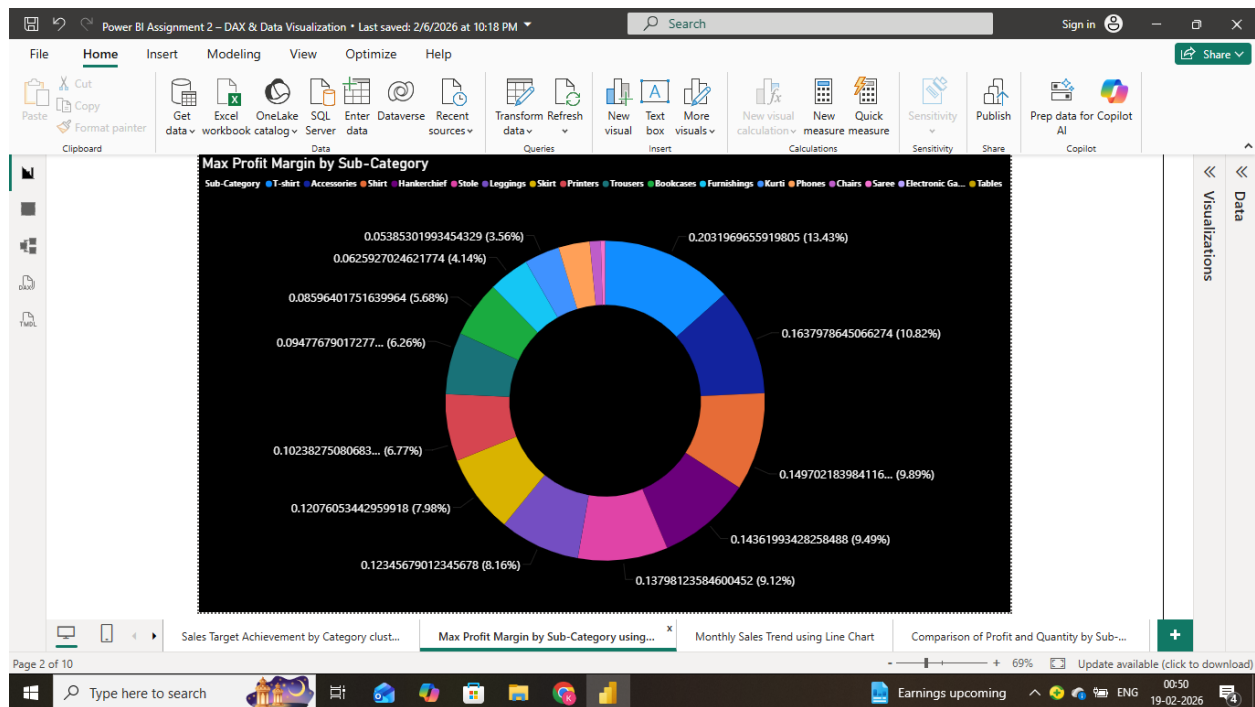
Data Visualization:

Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.

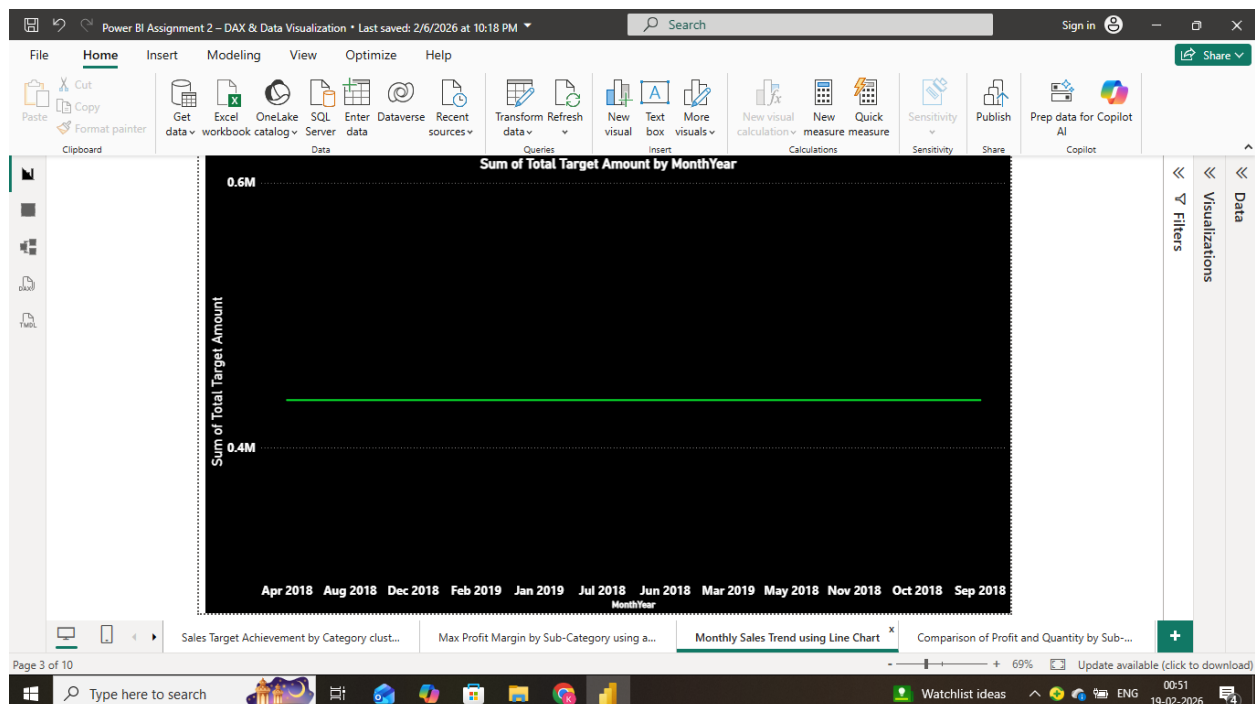


Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each

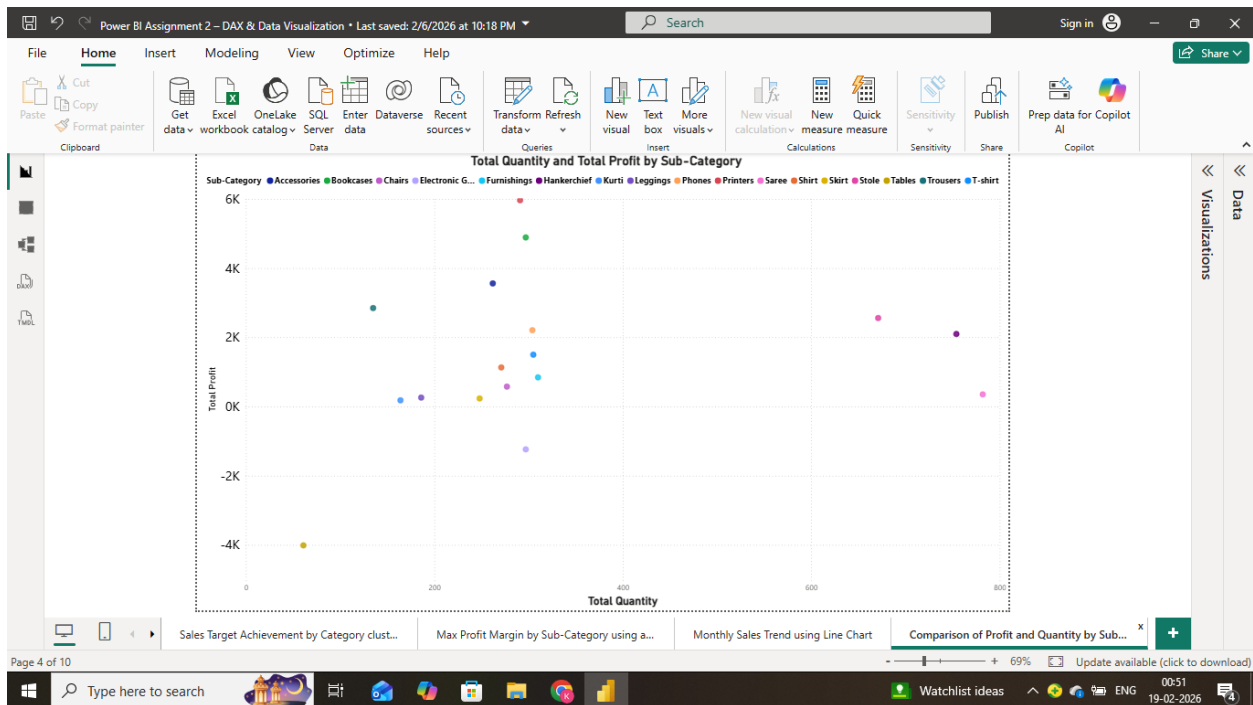
sub-category of products using a donut chart.



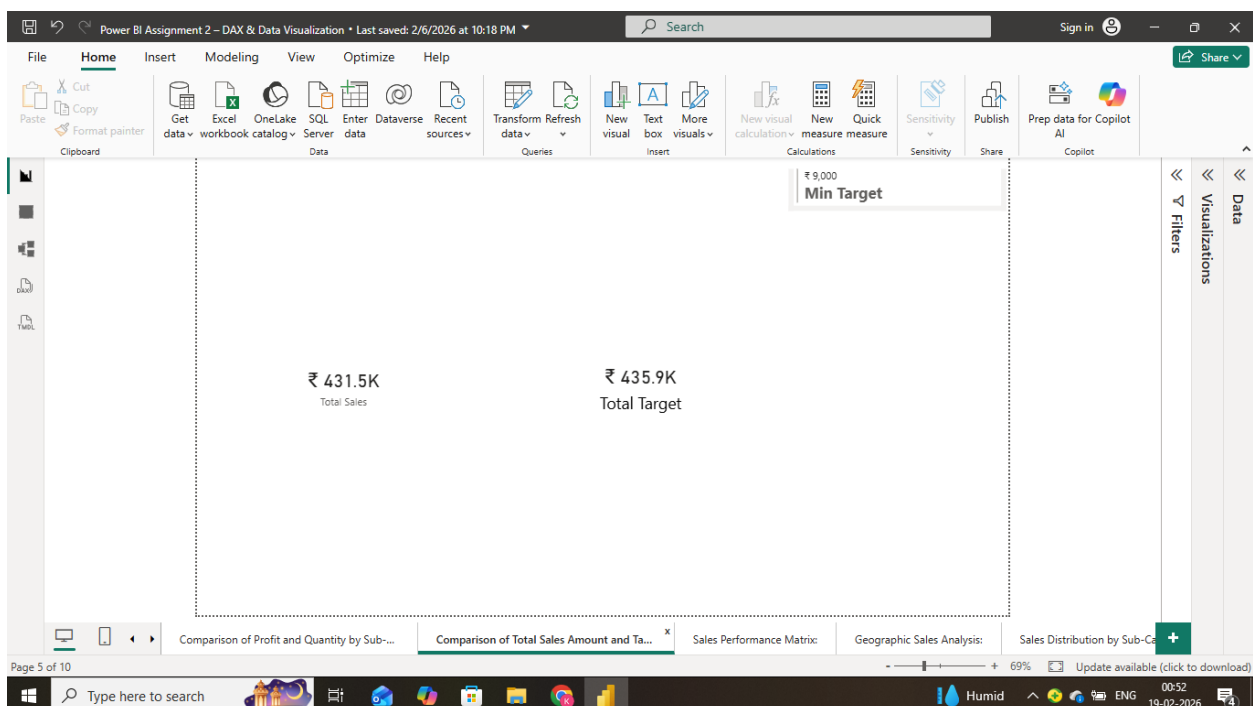
Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



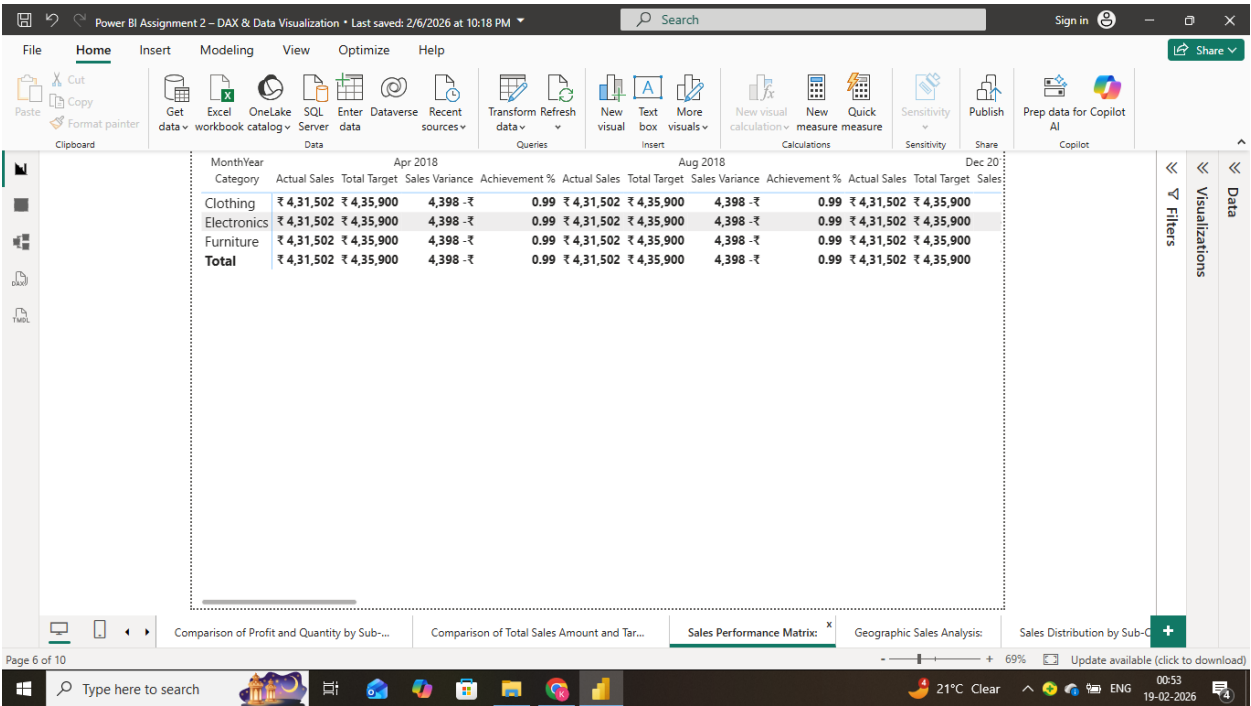
Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



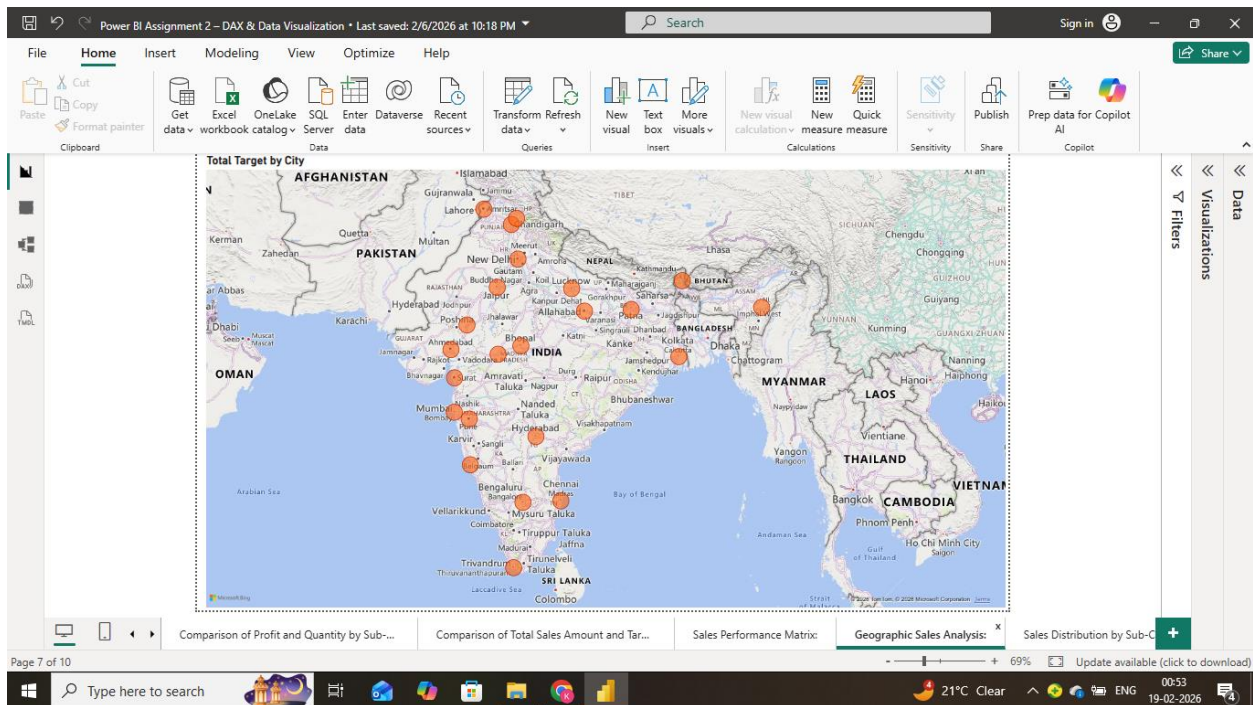
Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



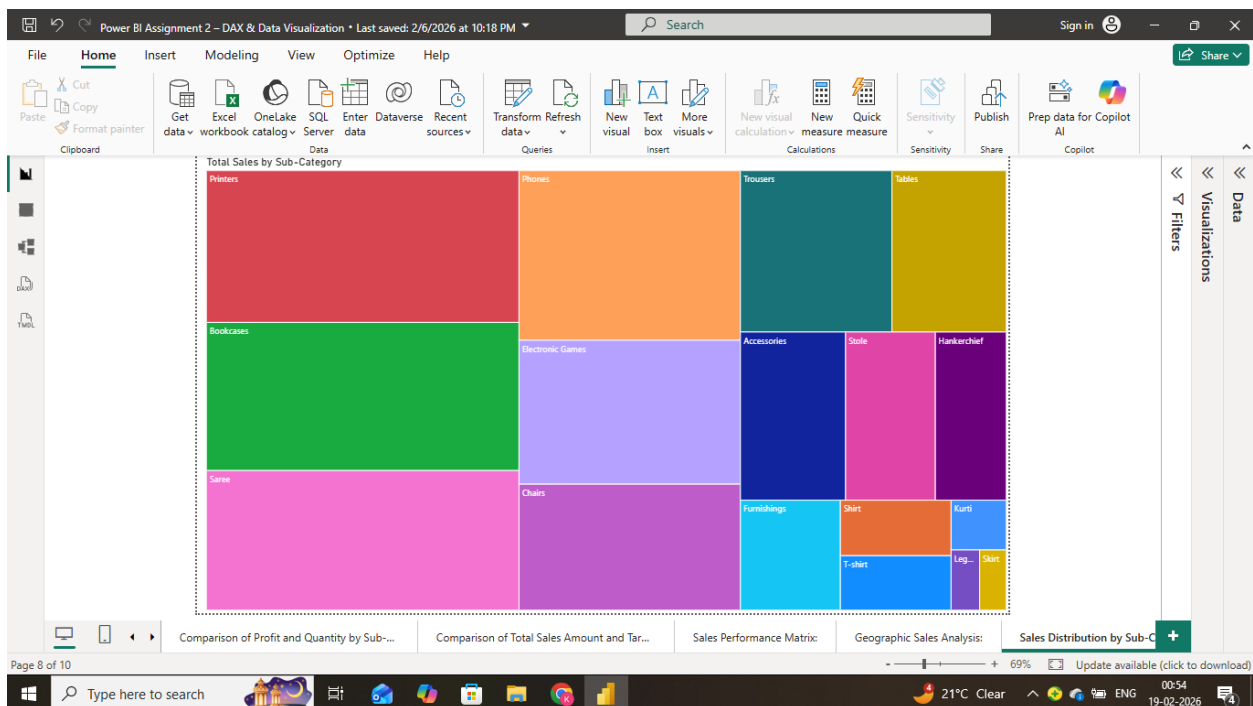
Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a treemap.



Order Count Analysis by State: Create a funnel chart to visualize the distribution of order

counts across different states.

